

CSMProtonics0000000001 — Protonic Communication: Future Systems Design

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMProtonics0000000001 – Protonic Communication: Future Systems Design V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Nafion proton conductivity	$\sigma = 0.082$ S/cm (literature estimate)	$\sigma = 0.092$ S/cm (2025 Meredith Instruments Nafion 115 characterization)	12% higher conductivity changes proton mobility and device speed calculations	Meredith Instruments 2025 characterization
H-FET gate voltage	2.5 V threshold	1.8 V threshold (ultra-thin 50 nm Nafion gate membrane)	Lowers power requirement; enables operation from piezoelectric harvested supply (1.8 V OPEN circuit)	2025 H-FET device optimization; Drexel 2025
Gate electrode material	Au/Pd alloy (palladium critical mineral)	Ti ₃ C ₂ T _x MXene gate electrode (no Pd, work function 4.7 eV compatible)	Pd is a critical mineral with supply risk; MXene eliminates this dependency	Nature Electronics 2025

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Proton conductor at high temp	Nafion only (requires humidity, max ~120°C)	BaZrO ₃ :Y ceramic proton conductor added (200-600°C, solid-state)	Post-Carrington fire scenarios require proton conductor beyond Nafion thermal limits	Journal of Materials Chemistry A 2025
Propagation latency	45 µs/mm (rough estimate)	2.8 µs/mm (calculated from 2025 measured mobility data)	16× improvement changes feasibility assessment for several applications	Nafion 115 characterization 2025
System architecture	Standalone protonic logic gates	Hybrid protonic-photonic: optical fiber long-haul + H-FET endpoints	Pure protonic too slow for medium-speed data; hybrid achieves GHz photon bus + GIC-immune node logic	Practical architecture optimization 2025
Power supply for H-FET	External electronic power assumed	Piezoelectric KNbO ₃ -BaTiO ₃ harvester providing 1.8 V gate bias (lead-free)	GIC-immune system must also have GIC-immune power supply; piezo harvesting closes the loop	TDK 2025; Nano Energy 2024
Solar Cycle 25 urgency	Generic electromagnetic threat framing	SSN ~137 peak; current decade highest GIC risk in 60+ years — deployment window is NOW	Risk is present not future; implementation timeline must reflect current solar activity	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
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System architecture	Pure protonic too slow for medium-speed data; hybrid achieves GHz photon bus + GIC-immune node logic
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Solar Cycle 25 urgency	Risk is present not future; implementation timeline must reflect current solar activity

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMProtonics0000000001 — Protonic Communication: Future Systems Design V1.0 archived; V2.0 is the active specification as of June 2026.